

PRAHARI

AI-Powered Digital Public Safety Command Centre

Predict. Protect. Prosecute.

ET AI Hackathon 2.0 — Phase 2: Build Sprint — Prototype Submission

Problem Statement #6: AI for Digital Public Safety —
Defeating Counterfeiting, Fraud & Digital Arrest Scams

1. The Problem: An Industrial-Scale Assault on Digital Trust

India is facing two converging threats to public trust in its financial and digital systems. The first is the **digital arrest scam**: organised, cross-border fraud compounds impersonate CBI, police, RBI and Enforcement Directorate officials over video calls — using spoofed numbers, AI-generated voices, deepfake video and forged government credentials — to hold victims in psychological custody for hours and coerce them into transferring their savings to 'verification accounts'. The Ministry of Home Affairs reported **over Rs 1,776 crore defrauded in just the first nine months of 2024**, against a backdrop of 1.14 million cybercrime complaints in 2023 (up 60% year-on-year) — with senior citizens disproportionately targeted.

The second is **Fake Indian Currency Notes (FICN)**: high-quality counterfeit Rs 500 notes routinely bypass manual detection at local cash counters, small businesses and even bank branches, entering circulation through known border corridors. Detection today depends on touch, instinct and expensive fixed machines that never reach the last-mile cash economy.

The deeper failure is structural: **every response today is reactive and siloed**. An FIR is filed after the money is gone; a counterfeit is discovered after it has changed hands; and the coordinated campaign behind hundreds of 'unrelated' cases across states remains invisible because no system connects victim reports, call infrastructure, device fingerprints and mule accounts into one picture.

2. The Solution: PRAHARI

PRAHARI (Hindi: 'sentinel') is a unified AI intelligence platform for law-enforcement agencies, financial institutions and citizens that shifts digital public safety from **reactive case investigation to predictive threat neutralisation**. One command centre, three AI engines — delivered through five operational modules in the prototype:

Module	Who it protects	What it does
Scam Shield NLP scam classifier	Citizens, telecom providers	Classifies live call transcripts against digital-arrest playbooks in under two seconds and alerts the victim before the transfer occurs , with automated telecom spoof-block requests and MHA/I4C incident reports.
FICN Scanner Computer vision	Bank tellers, merchants, field officers	Turns any smartphone camera, counting machine or PoS terminal into a currency authenticator — verifying seven RBI security features and cross-checking serial numbers against the national seizure registry.
Network Intel Graph AI	Police cyber cells, I4C, banks	Fuses victim reports, VoIP signatures, device fingerprints and mule-account linkages into one graph; community detection exposes the coordinated campaign across jurisdictions and exports a SHA-256-sealed, court-ready evidence package.

Two further modules put these engines to work at the edges: **Live Intercept**, a real-time fusion console where speech AI (synthetic-voice detection), vision AI (deepfake analysis), spoof-signature tracing and script classification converge on an active call and trigger intervention inside the pre-transfer window; and **Citizen Shield**, a WhatsApp-style conversational triage assistant that gives citizens explainable

fraud verdicts in English and Hindi (12-language roadmap) with guided reporting to the 1930 helpline. All engines feed a **National Command Centre**: a live geospatial threat map of fraud hotspots, scam-call origin zones and FICN entry corridors, with a real-time fused intelligence feed for command staff.

3. System Architecture

The prototype is a client-side React application; the architecture below shows the production data flow the prototype models end-to-end.

INGEST	AI ENGINES	ACTION
Telco call metadata & live transcripts (SIP/STT)	Scam Shield — weighted multi-category NLP signal fusion (authority impersonation, victim isolation, urgency/fear, financial extraction, identity harvesting); LLM-augmentable	Victim push-alert · telecom spoof-block · auto-generated MHA/I4C report
Smartphone / counting-machine / PoS image capture	FICN Scanner — CV verification of security thread, watermark & electrotpe, microlettering, intaglio relief, see-through register, latent image, serial-pattern validation	Instant teller verdict · seizure report pre-filled for NCRB FICN registry · batch/corridor linkage
I4C victim reports · CDR metadata · bank device fingerprints · account linkages	Network Intel — entity resolution, Louvain community detection, temporal correlation across 38+ entity graph	Campaign confirmation · mule-account freeze drafts · inter-state alerts · SHA-256-sealed evidence package

Technology stack

- **Frontend / command centre:** React 18 + Vite, Leaflet geospatial mapping, canvas-based force-directed graph rendering.
- **Scam classification (real, live):** weighted signal-fusion NLP engine over five behavioural categories derived from I4C-documented scam playbooks — every verdict in every module is computed live from the text; nothing is pre-scored.
- **Real audio pipeline:** neural TTS (Microsoft edge-tts, Indian-English voices, word-boundary sync) for the playbook call; OpenAI Whisper word-level transcription for REAL scam robocalls sourced from FTC enforcement evidence (NCSU robocall-audio-dataset, public domain); WebAudio AnalyserNode live spectrum; the intervention pauses the actual audio mid-sentence.
- **Live speech-to-text:** Web Speech API microphone mode — a judge can speak a scam line and watch the classifier react word by word.
- **Counterfeit detection (real inference, in-browser):** custom CV rectifier (OTSU segmentation, min-area-rect, homography warp) + MobileNet v2 embeddings + k-NN, trained on a public real-vs-fake note corpus plus the team's own note photographs. Hold-out accuracy: 5/5 genuine-vs-counterfeit on images the model never saw.
- **Graph intelligence:** typed entity graph (kingpin, operator cells, VoIP blocks, SIM boxes, device fingerprints, mule accounts, victims) with community detection and evidence-package export sealed by SHA-256 (Web Crypto).
- **Integrity & auditability:** every exported intelligence package carries an integrity seal and chain-of-custody metadata for legal admissibility.

4. Prototype Walkthrough (What the Demo Shows)

- **Command Centre:** national threat map with live hotspots (Jamtara, Nuh, Malda corridor, metro clusters) and a fused real-time intelligence feed.
- **Live Intercept (REAL recording):** an actual scam robocall from FTC enforcement evidence — 'your Social Security number will be suspended... a non-bailable arrest warrant' — plays audibly while Whisper-timed words stream into the classifier; the fusion engine crosses threshold at T+15s and audibly cuts the call, with no hand-tuned trigger, before the victim would reach the human extractor.
- **Live Intercept (Indian playbook):** the I4C digital-arrest script voiced with neural TTS (the same class of voice tooling fraud compounds deploy) is intercepted at the RTGS 'safe account' money ask — victim alert in Hindi and English, bank transaction hold, telecom spoof-block, auto-generated I4C case.
- **Live Mic:** real speech-to-text on the presenter's own voice, classified word-by-word as it is spoken.
- **Scam Shield:** a real digital-arrest script pattern is analysed forensically — risk score 97/100, verdict 'DIGITAL ARREST SCAM DETECTED', every manipulative phrase highlighted by signal category; a genuine bank service call scores near zero, demonstrating the low false-positive behaviour required of citizen-facing tools.
- **FICN Scanner (real inference):** photographs of actual notes are auto-cropped, perspective-corrected, embedded with MobileNet v2 and classified genuine-vs-fake in the browser — 5/5 correct on held-out images the model never saw, with denomination recognition; a guided RBI security-feature walkthrough remains as the educational tab.
- **Network Intel:** twelve seemingly unrelated victim reports across seven states are correlated into a single confirmed campaign ('OPX-2231 — Operation Parcel Trap', confidence 94%) 31 hours after the first report, with Rs 4.2 crore of flow mapped through six mule accounts and a court-ready evidence package exported in one click.
- **Citizen Shield:** a WhatsApp-style assistant triages forwarded scam descriptions in under two seconds with explainable verdicts (e.g. HIGH RISK 74/100 for a digital-arrest description, LOW RISK for a genuine bank call), switchable between English and Hindi, with one-tap 1930 report filing.

5. Mapping to the Evaluation Focus

Evaluation criterion	How PRAHARI addresses it
Digital arrest scam detection precision & recall	Weighted fusion across five independent signal categories — no single keyword can trigger a critical verdict; benign calls demonstrably score low. Categories map to documented I4C playbook stages, giving explainable verdicts.
Counterfeit detection across denominations & print quality	Per-feature verification with individual confidence scores generalises across the RBI security-feature set; serial-pattern validation catches even high-print-quality counterfeits that reuse or increment seized series.
Fraud network detection lead time	Graph correlation confirms a coordinated campaign at T+31h after the first victim report — while the campaign is still recruiting victims — versus months of siloed, per-FIR investigation.
Very low false positives for citizen-facing tools	Saturating category scores, verdict tiers and explainable per-signal breakdowns; the demo explicitly includes a genuine bank call as a negative control.
Auditability / legal admissibility	Evidence packages are SHA-256 integrity-sealed, timestamped, and carry chain-of-custody metadata (source systems, lawful-intercept references).

6. Impact & Why This Matters

- **Prevention, not post-mortem:** the single highest-leverage moment in a digital arrest scam is the window before the first transfer. Scam Shield is built for exactly that window.
- **Last-mile currency trust:** FICN detection moves from expensive fixed infrastructure to every smartphone — reaching kirana stores, petrol pumps and rural bank correspondents.
- **From cases to campaigns:** Network Intel gives police what fraud compounds fear most — cross-jurisdiction visibility, fast enough to freeze mule accounts while money is still in the pipeline.
- **Citizen reach:** the roadmap extends Scam Shield to WhatsApp/IVR in 12 regional languages, guided reporting to the 1930 helpline and NCRB portals.

7. Production Roadmap

- **Phase 1 (0–3 months):** pilot Scam Shield with one telecom circle using on-device speech-to-text; fine-tune transformer classifier on I4C complaint corpus; FICN model training on RBI/NCRB seizure imagery.
- **Phase 2 (3–9 months):** bank integrations (counting machines, teller apps); I4C complaint-pipeline ingestion for Network Intel; evidence-package format alignment with BNSS digital-evidence rules.

- **Phase 3 (9–18 months):** national command-centre deployment, deepfake/AI-voice detection module, 12-language Citizen Shield on WhatsApp and IVR.

Note: all statistics displayed inside the prototype UI, and all case entities, are synthetic demonstration data modelled on publicly reported I4C/NCRB patterns. The currency specimen is a stylised SVG render, not an image of real currency. Public figures cited in Section 1 are from press reporting of I4C data.

PRAHARI — built for ET AI Hackathon 2.0, Phase 2 Build Sprint. Repository and 3–4 minute demo video accompany this document.